

Exact Hat Algebra and the Gains from Trade

International Trade

Farid Farrokhi

Introduction

- We are often interested in running “comparative statics analyses”

What are the effects of a policy as changes to exogenous variables (e.g. endowments, technology, trade costs) on endogenous variables (e.g. trade flows, prices, wages, welfare)?

- GE trade models often incorporate many parameters, including elasticity parameters, demand shifters, supply shifters, and trade costs.
- In practice, it can be too hard to estimate ALL parameters of a model.
- It turned out that it might be also unnecessary!
- The method of Exact Hat Algebra: We do not need to know all model parameters to compute the effect of a policy on equilibrium outcomes.
- We illustrate the basics of this method, using the Armington model. Again, the method is applicable to many more advanced models.

Recap: Armington Model

- Price index:

$$P_j = \left[\sum_{\ell=1}^N b_{\ell j} d_{\ell j}^{1-\sigma} (w_{\ell}/a_{\ell})^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \quad (1)$$

- Trade shares:

$$\pi_{ij} = \frac{b_{ij} d_{ij}^{1-\sigma} (w_i/a_i)^{1-\sigma}}{P_j^{1-\sigma}} \quad (2)$$

- Market clearing condition:

$$w_i L_i = \sum_{j=1}^N \pi_{ij} w_j L_j \quad (3)$$

- Welfare:

$$U_i = \frac{w_i L_i}{P_i} \quad (4)$$

- Eqs (1)-(4) take $\{L_i, a_i, b_{ij}, d_{ij}, \sigma\}$ as given to predict $\{P_i, \pi_{ij}, w_i, U_i\}$.

Exact Hat Algebra

- Consider a policy as a change to exogenous variables.
- For any generic variable x in the “baseline” equilibrium, let x' be its corresponding value in the post-policy (“new”) equilibrium, and

$$\hat{x} \equiv \frac{x'}{x}$$

- To fix the idea, let us consider a change to trade costs $\mathbf{d} = [d_{ni}]$ to $\mathbf{d}' = [d'_{ni}]$.
- We proceed to write the equilibrium in terms of hat variables!

Exact Hat Algebra: Price index

- Price index

$$(P')_j^{1-\sigma} = \sum_{n=1}^N b_{nj} \left(d'_{nj}\right)^{1-\sigma} (w'_n/a_n)^{1-\sigma}$$

$$P_j^{1-\sigma} = \sum_{\ell=1}^N b_{\ell j} d_{\ell j}^{1-\sigma} (w_{\ell}/a_{\ell})^{1-\sigma}$$

- Therefore,

$$\begin{aligned}\hat{P}_j^{1-\sigma} &= \sum_{n=1}^N \frac{b_{nj} \left(d'_{nj}\right)^{1-\sigma} (w'_n/a_n)^{1-\sigma}}{\sum_{\ell=1}^N b_{\ell j} d_{\ell j}^{1-\sigma} (w_{\ell}/a_{\ell})^{1-\sigma}} \\ &= \sum_{n=1}^N \frac{b_{nj} d_{nj}^{1-\sigma} (w_n/a_n)^{1-\sigma}}{\sum_{\ell=1}^N b_{\ell j} d_{\ell j}^{1-\sigma} (w_{\ell}/a_{\ell})^{1-\sigma}} \hat{d}_{nj}^{1-\sigma} \hat{w}_n \\ &= \sum_{n=1}^N \pi_{nj} \hat{d}_{nj}^{1-\sigma} \hat{w}_n\end{aligned}$$

- Eq (1) in hat algebra:

$$\hat{P}_j = \left[\sum_{n=1}^N \pi_{nj} \hat{d}_{nj}^{1-\sigma} \hat{w}_n \right]^{\frac{1}{1-\sigma}}$$

Exact Hat Algebra: Trade shares

- Trade shares in the baseline and new equilibria

$$\pi'_{ij} = \frac{b_{ij} (d'_{ij})^{1-\sigma} (w'_i/a_i)^{1-\sigma}}{(P'_j)^{1-\sigma}}$$

$$\pi_{ij} = \frac{b_{ij} d_{ij}^{1-\sigma} (w_i/a_i)^{1-\sigma}}{P_j^{1-\sigma}}$$

- Eq (2) in hat algebra:

$$\hat{\pi}_{ij} \equiv \frac{\pi'_{ij}}{\pi_{ij}} = \frac{\frac{b_{ij} (d'_{ij})^{1-\sigma} (w'_i/a_i)^{1-\sigma}}{(P'_j)^{1-\sigma}}}{\frac{b_{ij} d_{ij}^{1-\sigma} (w_i/a_i)^{1-\sigma}}{P_j^{1-\sigma}}} = \frac{\hat{d}_{ij}^{1-\sigma} \hat{w}_i^{1-\sigma}}{\hat{P}_j^{1-\sigma}}$$

- Note: All endogenous variables in the new equilibrium take the form of x' . Among exogenous variables, only the ones affected by the policy must take the form of x' .

Exact Hat Algebra: MCC and Welfare

- Labor MCC in the new equilibrium

$$w'_i L_i = \sum_{j=1}^N \pi'_{ij} w'_j L_j$$

which can be expressed as:

$$\widehat{w}_i w_i L_i = \sum_{j=1}^N \widehat{\pi}_{ij} \pi_{ij} \widehat{w}_j w_j L_j$$

- Welfare:

$$\widehat{U}_i \equiv \frac{\frac{w'_i L_i}{P'_i}}{\frac{w_i L_i}{P_i}} = \frac{\widehat{w}_i}{\widehat{P}_i}$$

Exact Hat Algebra: System of GE

- Putting together, the GE in hat notation takes $\{\pi_{ij}, w_i L_i, \sigma\}$ and $\{\hat{d}_{ij}\}$ as given to predict $\{\hat{P}_j, \hat{\pi}_{ij}, \hat{w}_i, \hat{U}_i\}$

$$\hat{P}_j = \left[\sum_{n=1}^N \pi_{nj} \hat{d}_{nj}^{1-\sigma} \hat{w}_n^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \quad (5)$$

$$\hat{\pi}_{ij} = \frac{\hat{d}_{ij}^{1-\sigma} \hat{w}_i^{1-\sigma}}{\hat{P}_j^{1-\sigma}} \quad (6)$$

$$\hat{w}_i w_i L_i = \sum_{j=1}^N \hat{\pi}_{ij} \pi_{ij} \hat{w}_j w_j L_j \quad (7)$$

$$\hat{U}_i = \frac{\hat{w}_i}{\hat{P}_i} \quad (8)$$

- The system of Eqs. 5-8 is the counterpart of the system of Eqs. 1-4.
- $\{\pi_{ij}, w_i L_i, \sigma\}$ is **sufficient** to compute the GE changes in response to $\{\hat{d}_{ij}\}$.
 - In comparison, solving the initial and new GE in levels required $\{L_i, a_i, b_{ij}, d_{ij}, d'_{ij}, \sigma\}$.

A Numerical Algorithm

- Set tolerance, e.g., at 10^{-9} . Guess $\widehat{w}_i^{(1)} \forall i$ for iteration $t = 1$.
- At iteration $t \geq 1$,
 1. Normalize w.r.t. the numeraire (labor in country i_0): Replace $\widehat{w}_i^{(t)}$ with $\widehat{w}_i^{(t)} / \widehat{w}_{i_0}^{(t)}$.
 2. Calculate labor supply and demand for all i ,

$$\mathcal{S}_i^{(t)} = \widehat{w}_i^{(t)} w_i^{(t)} L_i, \quad \mathcal{D}_i^{(t)} = \sum_{j=1}^N \frac{\widehat{d}_{ij}^{1-\sigma} \widehat{w}_i^{1-\sigma}}{\sum_{n=1}^N \pi_{nj} \widehat{d}_{nj}^{1-\sigma} \widehat{w}_n} \widehat{w}_j w_j L_j$$

3. Calculate error $e = \max_i \left| \mathcal{D}_i^{(t)} / \mathcal{S}_i^{(t)} - 1 \right|$. If e is smaller than tolerance $= 10^{-9}$, convergence is achieved: Stop.
Otherwise, update the current guess of wage, $\widehat{w}_i^{(t)}$, to a new one, $\widehat{w}_i^{(t+1)}$, for all i

$$\widehat{w}_i^{(t+1)} = \left(\mathcal{D}_i^{(t)} / \mathcal{S}_i^{(t)} \right) \times \widehat{w}_i^{(t)}$$

and, take $\widehat{w}_i^{(t+1)} \forall i$ to iteration $t+1$.

Welfare

- Trade share of a country with itself (“domestic expenditure share”) can be expressed as:

$$\hat{\pi}_{ii} = \frac{\hat{d}_{ii}^{1-\sigma} \hat{w}_i^{1-\sigma}}{\hat{P}_i^{1-\sigma}} = \hat{d}_{ii}^{1-\sigma} \hat{U}_i^{1-\sigma}$$

- Hence,

$$\hat{U}_i = \hat{d}_{ii} \hat{\pi}_{ii}^{\frac{1}{1-\sigma}}$$

- Since $d'_{ii} = d_{ii} = 1$ (by normalization), we have also $\hat{d}_{ii} = 1$. Hence:

$$\hat{U}_i = \hat{\pi}_{ii}^{\frac{1}{1-\sigma}}$$

The gains from trade

- Change to welfare is entirely predicted by change to π_{ii} , with the extent of the relationship governed by σ .
- The gains from trade, as the absolute value of the percentage change in real income as we move from the observed equilibrium to autarky, equals:

$$\text{GFT}_n = \left| \frac{U_n^{\text{autarky}} - U_n^{\text{baseline}}}{U_n^{\text{baseline}}} \right| = 1 - \pi_{nn}^{\frac{1}{1-\sigma}} \quad (9)$$

- We ultimately care about welfare implications of a policy. All we need to know are baseline domestic expenditure share, π_{nn} , and the elasticity parameter, σ .
- We will appreciate this result even more if we step back and think about all complicated GE forces at work.